Cranial Prostheses for Chronic, Pan-Cortical Two-Photon Imaging in Behaving Mice

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Abstract

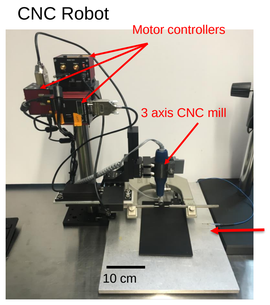
Chronic long term imaging of neurons has become increasingly important in recent years to understand and comprehend how information is encoded in the central nervous system at the single and population level. There have been several technological strides made to help neuroscientist visualize spontaneous and stimulated neuron firing, but have been limited to small regions of observation. To address the limitations of current chronic cranial imaging window preparations, we have developed 3D printed morphologically realistic, transparent cranial prostheses that allow cellular resolution optical access to the entire dorsal cortex. We have validated the ability to implant these 3D printed cranial implants for long durations, and preliminary experiments indicate a capacity to perform two-photon (2P) imaging in the chronically implanted Thy1-GCaMP6f mouse and detect Ca^{2+} activity in individual neurons. The implanted mouse was trained in motorized disc treadmill, while simultaneously carrying out *in vivo* optical 2P imaging. Future work includes extending 2P to the whole dorsal cortex and histological analysis to assess the inflammatory response of chronic implantation. In addition, we plan to use this technology to learn mesoscale and cellular scale activity of defined cell populations in an animal model. The tottering mice (*tg/tg*) are a mouse model used to study Episodic Ataxia type 2, a disease caused by a mutation in the CACNA1A gene and exhibit several characteristics, including transient motor ataxia, which can be triggered by stress, caffeine, and alcohol.

Approach

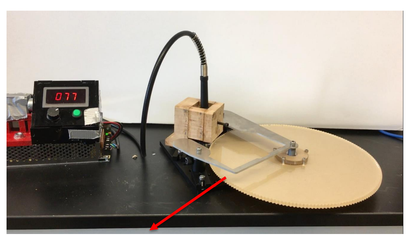
Design and fabrication

Scan the brain → Interpolated model → Render 3D surface → 3D CAD model → 3D printed prosthesis

Experimental Set up

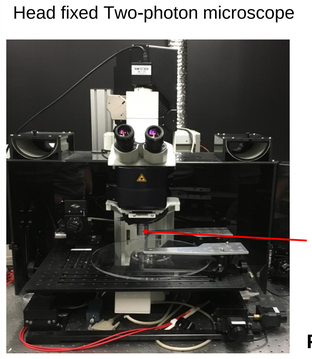


Standard rodent stereotax



Motorized disc treadmill for head fixed awake recording

Head fixed Two-photon microscope



Microscope objective

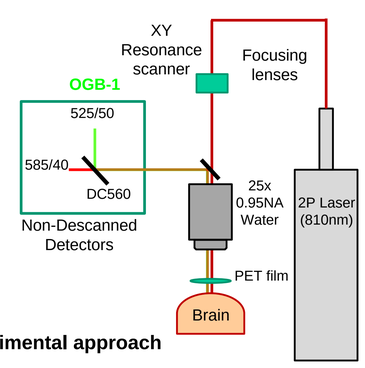
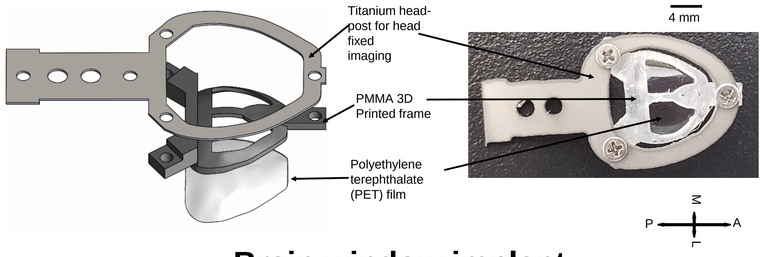


Figure 1. Experimental approach

Results

Composite brain window



Brain window implant



Figure 3. A step by step image illustrating the chronic brain window implant procedure. These stages ensure that the brain window implant is secure and remains intact for extended period of time of time.

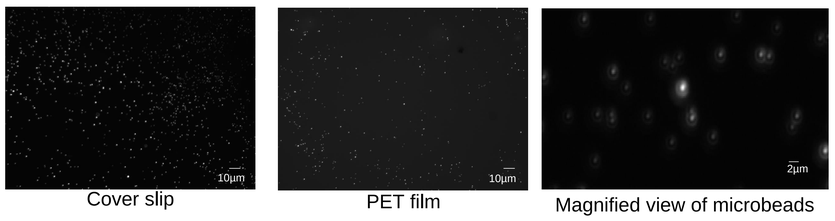
Cranial prosthesis allows long term optical access



Day 2 Day 14 Day 21

Figure 4. Brain window implant was imaged periodically to assess the quality and transparency of the brain post-surgery. There is limited to no regrowth of the skull and dura and which allow clear optical access for two-photon imaging.

Image quality analysis



Cover slip PET film Magnified view of microbeads

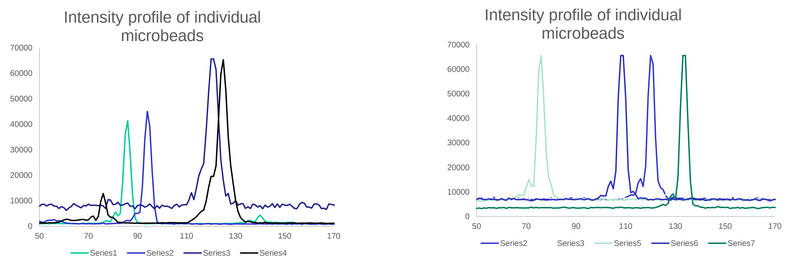


Figure 4. The quality of images produced by PET film was assessed and compared to cover glass brain window and PET film using point spread function (PSF) of microsphere beads. The result showed higher intensity in PET film compared to cover glass.

Wide-field *in vivo* two-photon Ca^{2+} imaging

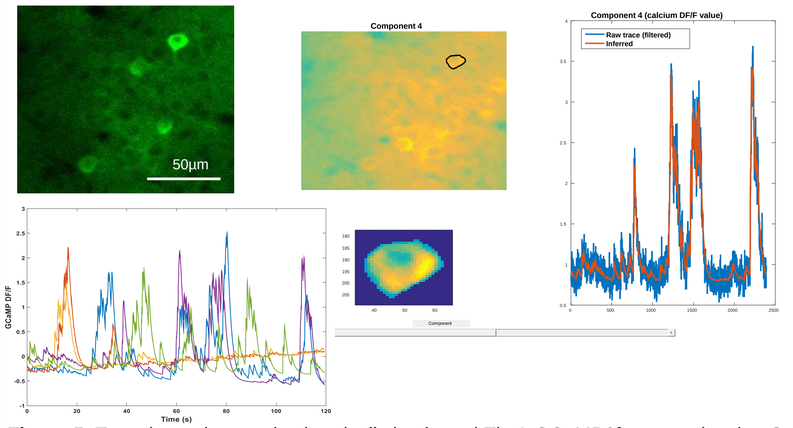


Figure 5. Two photon images in chronically implanted Thy1-GCaMP6f mouse showing Ca^{2+} activity. The Components are the individual cells that a MATLAB program has identified. The blue traces are the raw data, and the red traces, the output from the code. The large image to the left of the traces is a pseudocolored background image, and the small image below it is the spatial component of the selected cell. The image is DF/F traces

Discussion

- The new technology has demonstrated that we can image the brain over an extended period.
- The brain quality was well maintained, and all the bleeding was cleared immediately.
- The PET provided a transparent view of the brain which limited the dispersion of light and image quality was maximized.
- Another advantage of this method is that the materials are readily available and can be 3D printed.
- Future directions include histological analysis for microglial and astrocytes activation.
- Application of the new technology to disease animal models.

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